

Robot Framework — Task 5

1. Experiment Title

Test Setup and Teardown Using Robot Framework

2. Problem Statement

Set up a Robot Framework test suite with a **common test setup** that opens a browser and logs in before each test case. Define a **test teardown** that logs out and cleans up the browser resources after each test case.

3. Objective

To implement and demonstrate the use of **Test Setup** and **Test Teardown** in Robot Framework for common browser initialization, login, logout, and resource cleanup operations.

4. Tools, Software and Concepts Used

Tools / Software

- Python
- Robot Framework
- SeleniumLibrary
- Google Chrome
- Visual Studio Code

Concepts

- Test Setup
- Test Teardown
- Selenium WebDriver
- Browser automation
- Login automation
- Logout and resource cleanup
- Robot Framework variables
- User-defined keywords

- Open Browser
- Close All Browsers

5. Source Code / Implementation

File: test_setup.robot

*** Settings ***

Library SeleniumLibrary

Test Setup Open Browser And Login

Test Teardown Logout And Close Browser

*** Variables ***

\${URL} https://www.saucedemo.com/

\${USERNAME} standard_user

\${PASSWORD} secret_sauce

*** Test Cases ***

Verify Inventory Page

Location Should Contain inventory.html

Page Should Contain Products

Verify Products Page

Location Should Contain inventory.html

Page Should Contain Element css=.inventory_list

*** Keywords ***

Open Browser And Login

Open Browser \${URL} Chrome

Maximize Browser Window

Input Text id=user-name \${USERNAME}

Input Text id=password \${PASSWORD}

Click Button id=login-button

Location Should Contain inventory.html

Logout And Close Browser

Click Button id=react-burger-menu-btn

Wait Until Element Is Visible id=logout_sidebar_link 5s

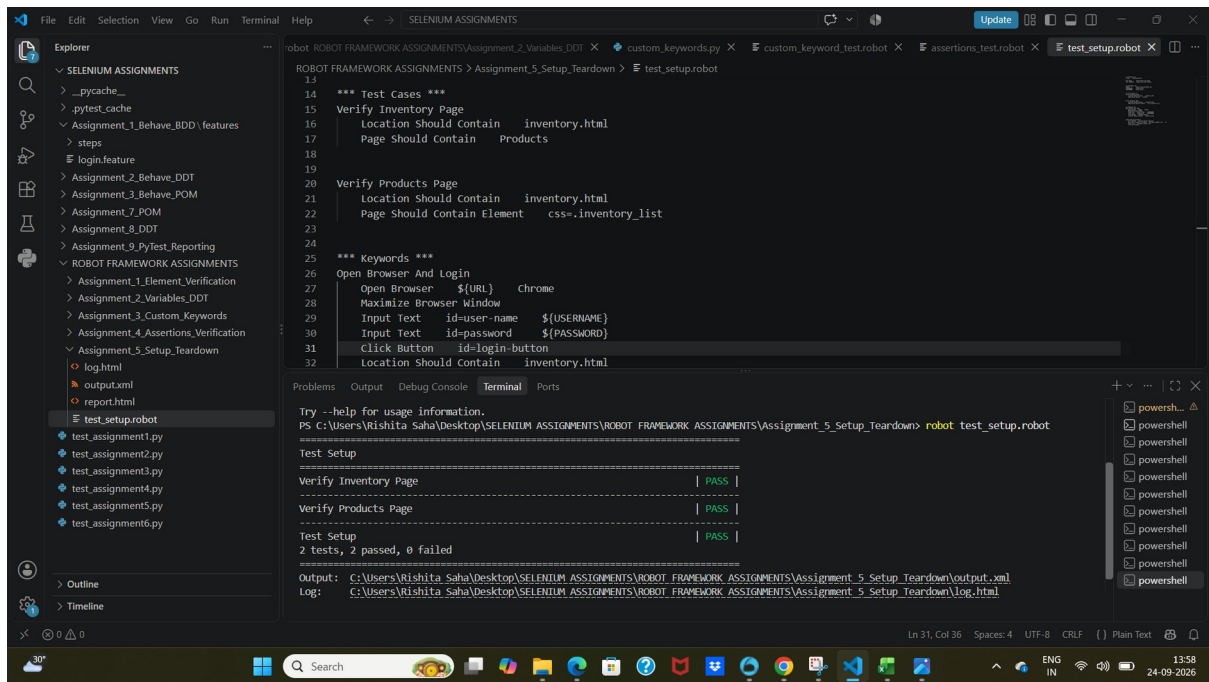
Click Link id=logout_sidebar_link

Close All Browsers

Implementation Steps

1. Import SeleniumLibrary.
2. Define Open Browser And Login as the common **Test Setup**.
3. Define Logout And Close Browser as the common **Test Teardown**.
4. Store the URL, username, and password as Robot Framework variables.
5. Before each test, open Chrome and log into SauceDemo.
6. Execute the individual test case.
7. After each test, open the menu, log out, and close the browser.
8. Execute two test cases using the same setup and teardown.

6. Output with Clearly Labelled Screenshots



Caption:

Robot Framework HTML report showing setup and teardown test execution

7. Result

The Robot Framework test suite successfully executed two test cases using a common browser setup and teardown. The browser was opened and the user was logged in before each test, and the browser was logged out and closed after each test.

8. Observation

The Test Setup and Test Teardown settings eliminated the need to repeat browser initialization, login, logout, and cleanup steps in every test case. The same setup and teardown were automatically applied to both test cases.

9. Conclusion

The experiment demonstrated the use of **Test Setup and Test Teardown in Robot Framework**. Common pre-test and post-test operations were successfully centralized and automatically executed for each test case, improving test organization and resource management.